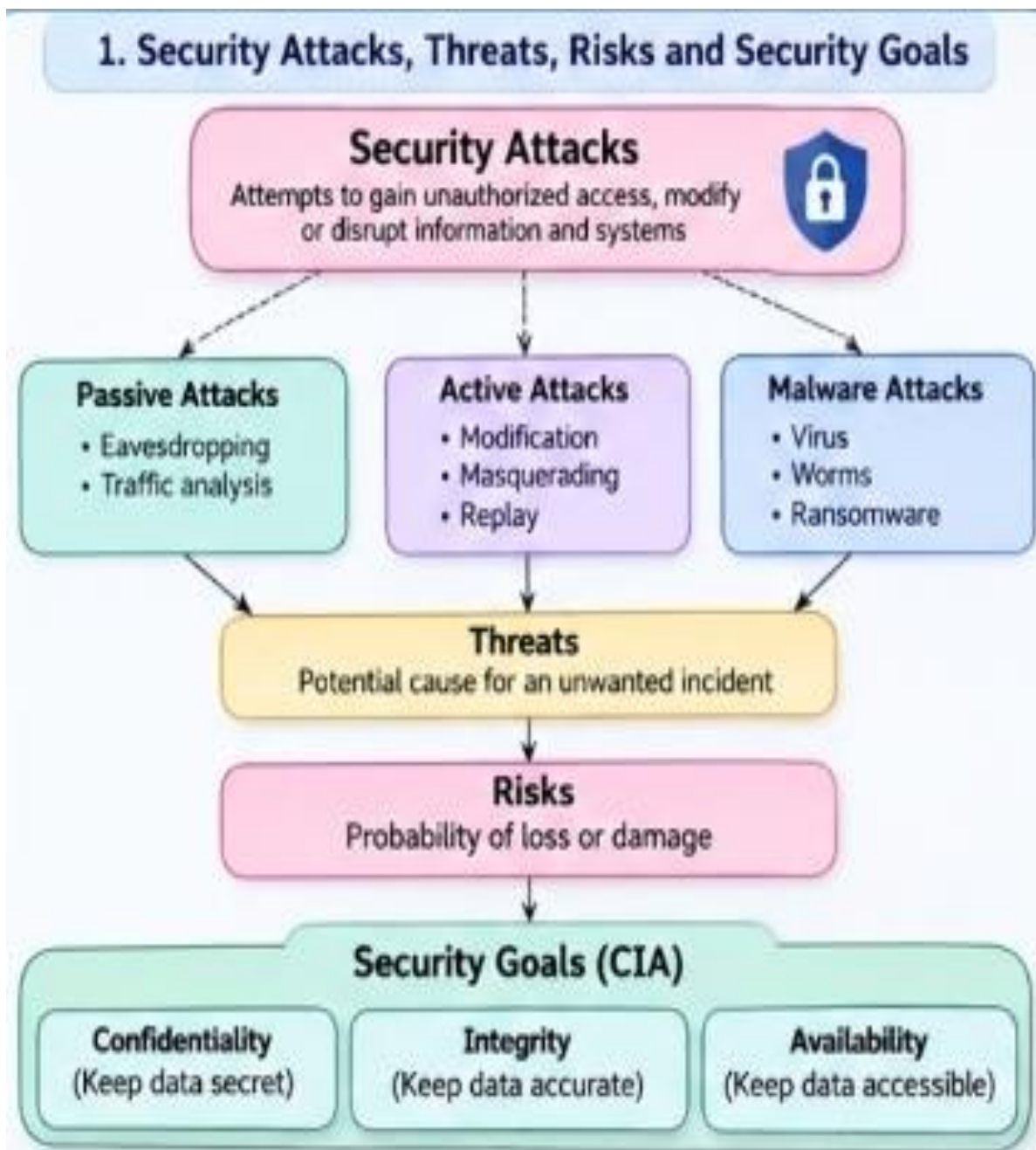
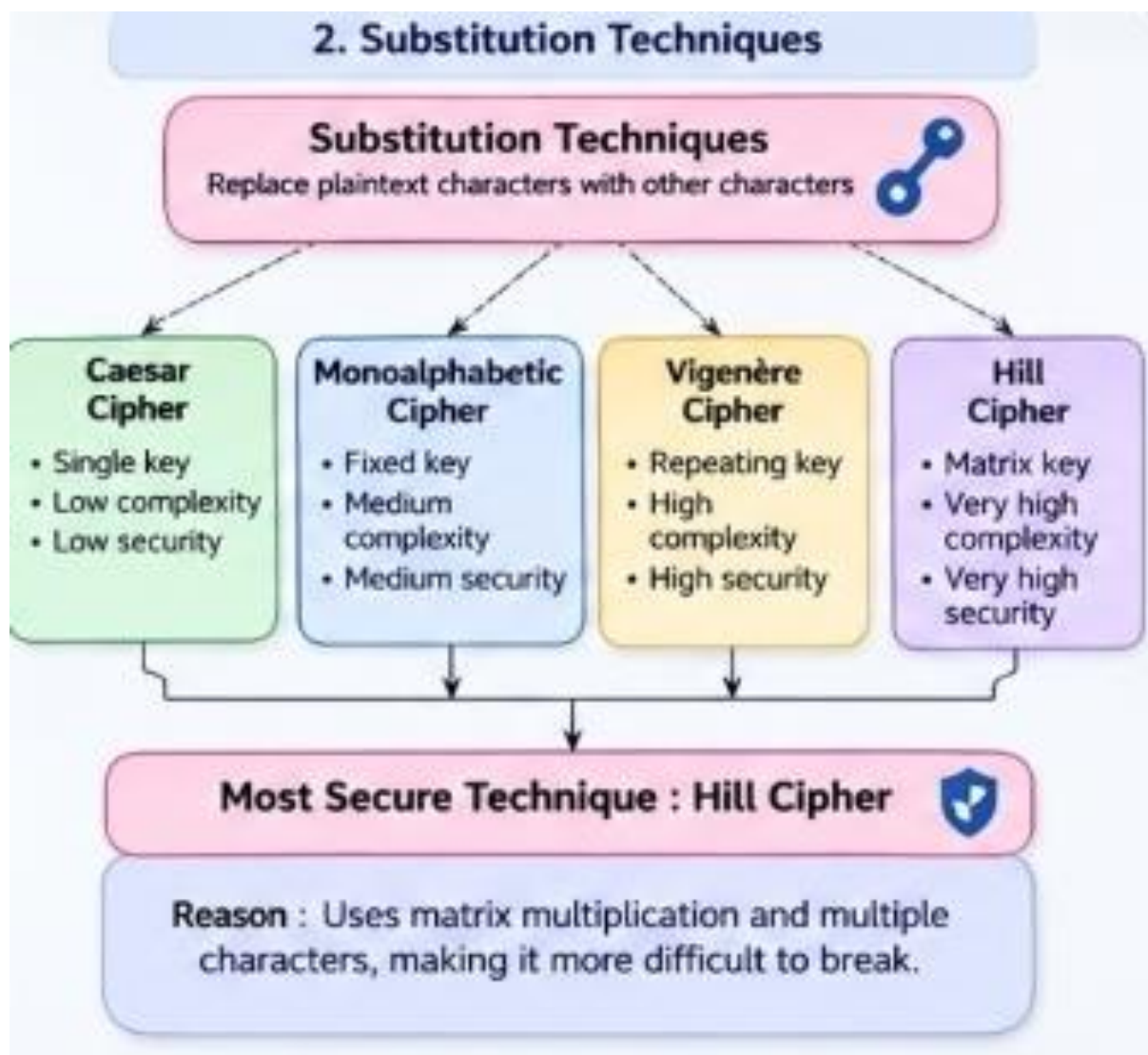


1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
- Show how each attack leads to specific threats and risks
 - Map how security goals help mitigate them Provide one real-world



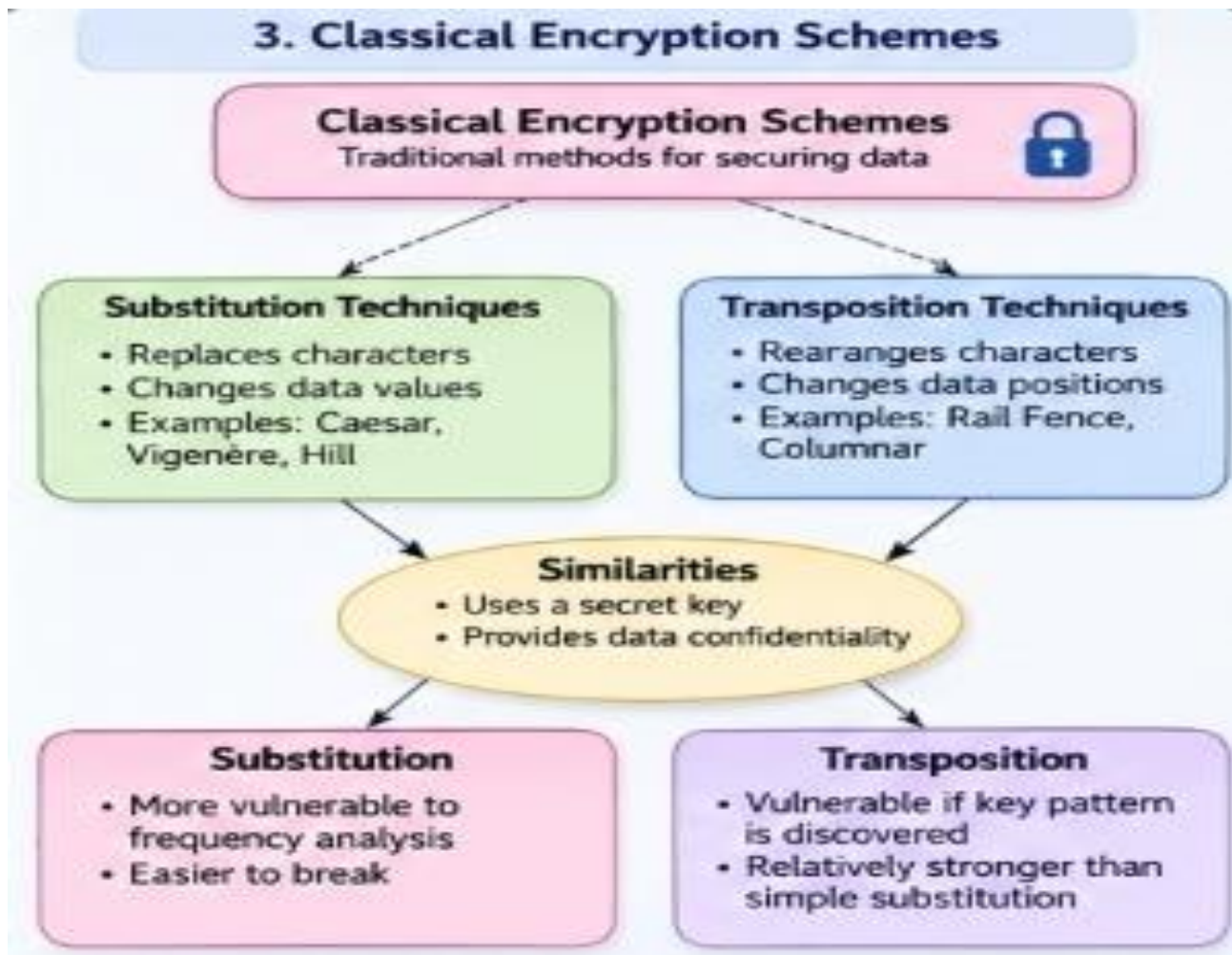
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks To connect based on: Key usage, Complexity and Security strength. Analyze which Technique is most secure and why.



3. Construct a concept map for Classical Encryption Schemes: Substitution

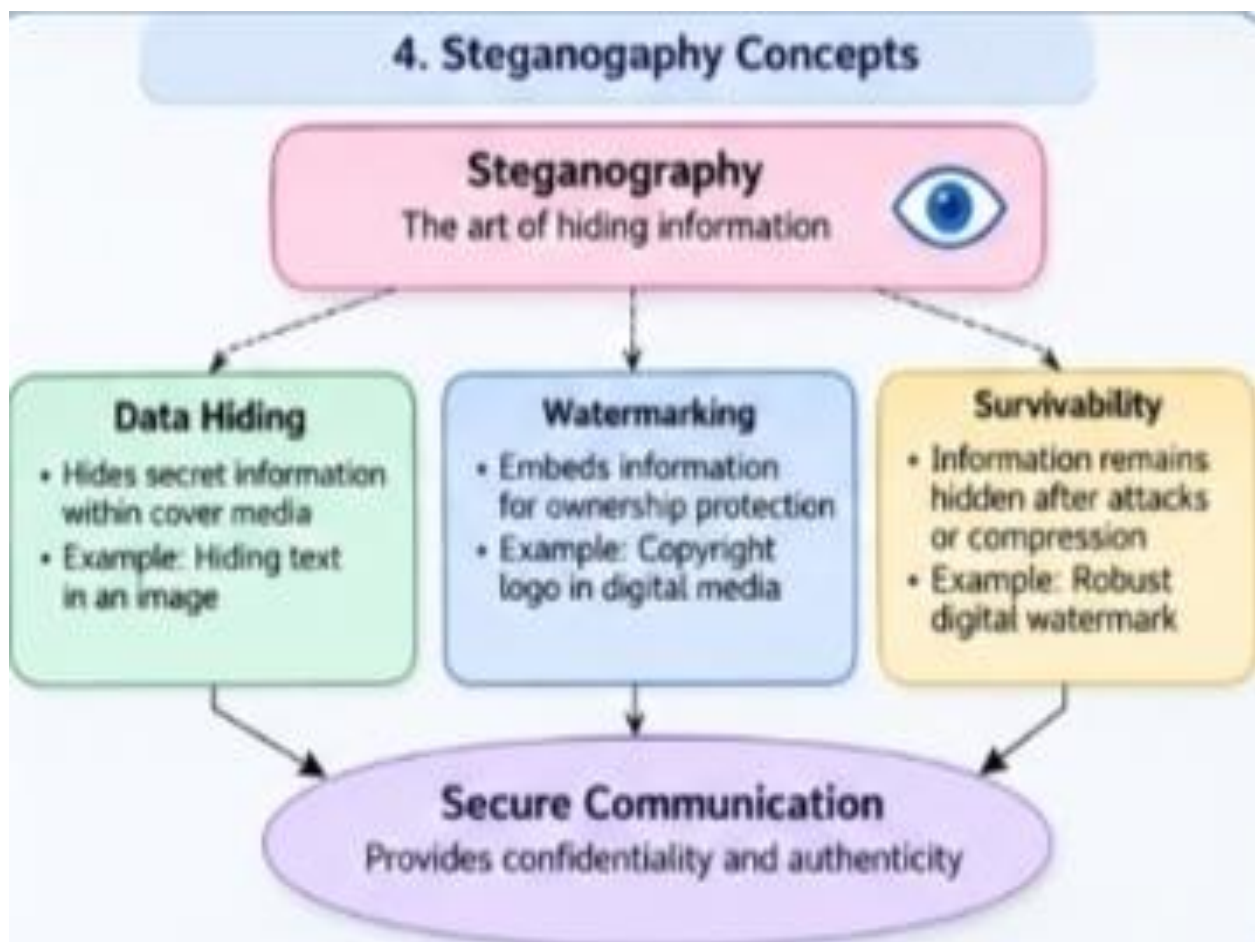
Techniques and Transposition Techniques. Perform the following tasks:

- a. Show differences and similarities
- i. Map how each affects the Data structure and Security level
- ii. Analyze which is more vulnerable to attacks and why



4.create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each



5. Draw a concept map linking number theory concepts used in cryptography:

Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi

Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography

